A SVM Binary Classifier for Automated Detection of Tomato Leaf Diseases in Low-cost Greenhouse Environments

by

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Abstract

Plant diseases significantly reduce agricultural yield, putting the global food supply at risk. Traditional pathogen detection methods rely on plant pathology expertise. Moreover, traditional methods are destructive, labor-intensive, and time-consuming. Thus, plant phenotyping image processing tools are used as an alternative for detecting plant pathogens; however, these tools can be inflexible, commercially owned, and expensive. Consequently, there is demand for flexible, reliable, open-source image processing tools for plant pathogen detection in low-cost agricultural environments. Accordingly, a support vector machine (SVM) binary classifier was developed to detect tomato leaf diseases. This paper presents and evaluates the performance of this image processing tool. The classifier is trained using textural parameters calculated from gray-level co-occurrence matrices. These parameters are extracted from images of tomato leaves in the publicly available PlantVillage dataset. Overall, the image processing tool is shown to be reliable and efficient in detecting tomato leaf diseases. The underlying implementation is open-source and utilizes modular computer vision libraries, enabling other researchers to modify or extend it. Additionally, the classifier can be turned into a mobile application for low-cost automated greenhouse management. Overall, the developed image processing tool is flexible, reliable, modifiable, and efficient in detecting tomato plant leaf pathogens.

Contents

Introduction

Plant diseases can have devastating effects on crops, leading to reduced overall yield [1, 2]. All approaches for developing disease-resistant crops or detecting plant pathogens require measurements of observable plant traits (phenotypes) [3]. Traditional plant phenotyping methods in plant pathogen detection require expertise. Moreover, these methods are destructive, labour-intensive, and time-consuming [4]. Therefore, traditional plant phenotyping methods are not suitable for monitoring large plant population sizes over time [5, 6, 7, 8]. As an alternative, non-destructive plant phenotyping methods using image processing have been developed [9], which are efficient, affordable, and allow larger samples of plants to be examined over time. However, these image-based solutions can be costly, making them inaccessible to less affluent plant growers [10, 11, 12]. Existing low-cost image-based phenotyping systems are patented, rendering the hardware and software of these systems inflexible and unmodifiable [13]. Furthermore, many image processing tools in plant phenotyping are designed for specific plant species and environments and therefore cannot be generalized to other plant species and contexts [6, 14]. Consequently, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing tools for plant phenotyping in the field of plant pathogen detection.

Recent developments in image-based plant phenotyping systems research show an increased development of plant phenotyping systems that use open-source hardware [15, 12] and software libraries [5]. Despite these advances, several challenges still remain. In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [12], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [11], and image compression techniques [16]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [17]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging;

however, their use is limited to the phenotyping of physiological plant traits [14]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [16]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [18].

In response to the aforementioned challenges of image-based phenotyping solutions in plant pathogen detection, this paper aims to develop a binary support vector machine (SVM) [19] classifier for the detection of tomato plant leaf diseases. The proposed classifier is trained using the publicly available PlantVillage dataset [20], and its overall reliability and performance are measured.

The remainder of this paper is organized as follows: Section 2, the literature review, presents the fundamentals of plant phenotyping and its relation to plant pathogen detection. Particular emphasis is placed on existing image-based systems for plant pathogen detection. Thereafter, in section 3, the methodology, we discuss the proposed classification framework, detailing how images are processed, how the SVM classifier is trained, and how the overall system's performance and reliability are evaluated. In section 4, we discuss our experimental results after our SVM classifier has been trained and evaluated. Lastly, in section 5, we discuss possible extensions of our classifier, potential future research directions, and implications of our experimental results.

Literature Review

The problem of plant diseases Our lives would not be possible without agriculture. We use goods like medicine, pesticides, and rubber everyday, all of which are made from plants. Plants are ubiquitous in our lives, and they fulfill some of our essential needs. Most importantly, plants feed most people globally [21]. But many of us still suffer from hunger. In 2024, it was estimated that between 638 and 720 million people did not have enough to eat [22]. Plant dieases make this situation even worse by robbing us of food [23, 24, 25, 20]. Therefore, they are a direct threat to our livlihoods.

Plant diseases make it difficult for crops to survive and stay healthy [26]. They interfere with a plant's cells causing them to malfunction or die [27]. And each cell helps the plant carry out essential life-sustaining functions: conducting photosynthesis, and absorbing water and nutrients. So, when these cells are damaged, the plant's ability to obtain and use nutrients is limited. As a result, plant diseases stunt the growth, and decrease the yield and nutritional value of crops [28, 29]. In some cases, the damage is so severe that farmers lose up to 100% of their crops [25, 30]. However, the losses caused by plant diseases goes further than just the yield and quality of crops; they extend to rural communities, consumers, the environment, and governments [28]. Therefore, plant diseases threaten the health of crops, and this has disastrous far-reaching effects on our livlihoods. Clearly, we need solutions to deal with them.

Some solutions to plant disease Luckily, several options exist. For example, plant breeding is an environmentally sustainable and cost-effective solution, which helps to produce disease-resistant varieties of plants [31, 32, 24, 9]. Alternatively, we also resort to precision agriculture, which uses data from crops and their environment to make optimal decesions about agricultural actions [33, 34, 35]. For instance, precision agriculture aids farmers in deciding on the best way to apply pesticides during a pathogen epidemic [36] by helping determine when to apply pesticides, how much pesticide to apply, and where to apply it; as a result, such precise pesticide application will reduce costs, save resources, and benefit the environment and human health [37, 38, 39, 40, 41]. So, plant breeding and precision agriculture help deal with plant diseases; however, these methods depend on effective plant phenotyping [5, 42, 3, 43, 44, 31].

The relationship between plant diseases and plant phenotyping Plant phenotyping is the evaulation of observable or measurable plant phenotypes [45, 46], which are proxy measurements or traits indicating a plant's stress, health or growth in relation to its environment [45, 14]. Moreover, phenotypes represent traits of an individual plant in terms of all of its chemichal, physiological, morphological, and behavioral aspects [47, 48, 14]. So, plant phenotyping provides quantative data that represents various aspects of a plant. When a plant is diseased its form, function and integrity are affected [26], therefore reliable measurments of all aspects of plant are required in plant disease solutions. Consequently, plant phenotyping is an effective quantative method for plant breeding and precision agriculture [3, 49, 30, 50, 31].

The relationship between plant phenotyping and plant breeding The goal of plant breeding is to develop plants with desirable traits like disease resistance. In this regard, the most important tasks for breeders are the selection of suitable individual plants and the developement of genetic material [31]. Plant phenotyping assists in these tasks. It does so by providing breeders data about a plant's phenotype-environment interactions, which allows them to identify and select suitable parameters for the development of improved crops [51]. And this method has been used successfully in several breeding programmes for disease resistance. To name a few, [52] conducted a study for Barley Rust Stripe resistance, [53] for Oat Crown Rust resistance, [54] for Wheat Head Blight resistance, [55] for Apple Scab resistance, and [56] for Cotton Verticillium Wilt resistance. Thus, plant phenotyping effectively assists plant breeding programmes.

The relationship between plant phenotyping and precision agriculture

2.1 Plant Phenotyping

Tradtional plant phenotyping solutions, and their challenges

Abiotic and biotic stressors limit cultivar biomass and yield [57], whilst human population growth strains the global food supply. Plant breeding, the process of inducing desirable, inheritable traits in plants through controlled human action [32], is now crucial in addressing the challenge of providing an adequate global food supply. Plant breeders require effective techniques to produce germplasm for cultivars that maximise yield and biomass under various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have proven to be an effective tool for plant breeding [31].

2.1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [45], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [46]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [47, 48, 14].

2.1.2 Plant Phenotyping As Method For Improving Plant Traits

Phenotypical traits serve as proxies in determining whether specific genotypes have favourable traits in particular environmental conditions. Quantitative data produced by plant phenotyping is most helpful in analyzing plant-environment interactions in plant breeding [45] for the development of germplasm with improved plant traits [14]. The data produced by phenotyping is used as a proxy for assessing a plant's performance in its environment. For example, stressors or disturbances in plants are induced by environmental conditions to which organisms are exposed, and these stressors induce symptoms that can be detected and evaluated using phenotyping. Hence, phenotypical traits of a plant have a complex dynamic feedback loop with their environment [58]. Many phenotypical traits are observable or quantifiable by image processing techniques.

2.1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

2.1.3.1 Simple Taxonomy of Phenotypical Traits

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [59]. Holistic properties consider the plant as a whole, whereas component-based features examine individual parts of the organism, such as its fruit, flower, leaves, stem, etc.

2.1.3.2 Leaf-Derived Phenotypical Traits: Robust Proxy Indicators of Plant Metabolic and Physiological Status

Leaf-derived traits are among the most used indicators of plant status because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth, and stress [7]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [60]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [61, 7], and leaf thickness for measuring abiotic stressors like water scarcity [62]. Additionally, physiological traits of leaves exist that indicate various factors of plant status, such as stomatal conductance for detecting biotic stressors like pathogens [63] and leaf water content for drought stress [64]. Considering the above, leaf-derived phenotypical traits are robust proxies of various metabolic and physiological plant parameters.

2.1.3.3 Alternative Phenotypical Traits

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [7]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [65], stems [66], seeds [67], and fruit [68].

2.2 Image-Based Plant Phenotyping

2.2.1 Image-Based Plant Phenotyping Versus Traditional Methods

Image-based plant phenotyping methods have several advantages over traditional methods. Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant's ontogenesis [6, 7, 8]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. These image processing solutions reduce the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [59]. Despite these advantages, several aspects of image processing, which determine the overall cost, efficiency, and applicability of image processing in plant phenotyping contexts, are important. The most important of these aspects are image acquisition and processing tasks.

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [68], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [69]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.2.2 Image Acquistion Techniques

2.2.2.1 The importance of Image Acquistion in Plant Phenotyping

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired, and the choice of imaging technique for data acquisition affects the types of phenotypical parameters that can be measured. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue, which are not visible to the naked eye [14]. Therefore, the appropriate selection of image acquisition techniques is crucial for analysing different types of plant traits.

2.2.2.2 Common Image Sensor Techniques In Plant Phenotyping

Commonly used image acquisition techniques include visible, fluorescence, thermal, tomographic, and hyperspectral imaging [70, 14]. The most popular techniques are visible and hyperspectral imaging due to their affordability, robustness and flexibility in measuring a wide range of phenotypic parameters. Visible imaging is an affordable option that utilises high-resolution RGB cameras. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [71, 72]. Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, allowing for detailed measurements of various phenotypic parameters. Notable applications of hyperspectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [73]. The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [14, 63], fluorescence imaging in plant pathogenic symptom monitoring [74, 75], and tomography in estimating plant organ water content [76].

2.2.3 Image Processing Techniques

2.2.3.1 The importance of Image Segmentation in Plant Phenotyping

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes, such as Arabidopsis [14, 77]. However, segmentation is a key image-processing task common to many image-based plant phenotyping solutions. Moreover, the validity and precision of the derived phenotypical measurements and analysis depend on the image segmentation method selected [17, 18], because the accuracy of more complex tasks, such as object detection, 3D reconstruction, and classification, depends on segmentation accuracy. Therefore, image segmentation methods are integral to image processing in plant phenotyping.

2.2.3.2 Commonly Used Image Segmentation Techniques

In plant phenotyping, selecting the appropriate segmentation technique is crucial when considering the type of phenotypic parameter to be measured. Segmentation methods are typically classifiable as traditional or learning-based [17]. Traditional segmentation methods typically target holistic phenotypical traits like plant height and include frame differencing [78], colour-based [79], edge detection [80], spectral band difference-based [81], shape modelling-based [82], and graph-based segmentation [83]. In contrast to traditional techniques, learning-based methods, which utilise machine learning or deep learning [84, 85], are more suitable for targeting component-based traits of specific organs or parts of a plant [17].

2.3 Low-cost Automated Image-Based Plant Phenotyping In Greenhouses

Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [86], automated conveyor belts, and benchtop configurations [87, 88], as well as advanced image sensors in high-throughput phenotyping systems [89]. However, these solutions are not accessible to all plant growers because they tend to be costly [10, 11, 12] Moreover, many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [13]. Furthermore, existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [6, 14]. Therefore, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing systems for plant phenotyping applications.

2.3.1 Open-source Software, Systems and Affordable Hardware As Cost Reducers In Image-Based Plant Phenotyping

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [15, 12] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and hyperspectral sensors to monitor phenotypic traits in greenhouse environments [11]. Additionally, there has been an increase in open-source image-processing plant phenotyping systems like Phenobox [13]. Despite these advances, several challenges remain in low-cost plant phenotyping systems.

2.3.2 Challenges In Automated Image-Based Plant Phenotyping

2.3.2.1 Large Image Sensor Data Throughput Neccesitates Robust Image Processing Tools

Modern image sensors in image-based phenotyping generate massive amounts of data [90], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [12]. Therefore, effective choices in image sensors are crucial for developing low-cost, automated image-based phenotyping systems [7].

2.3.2.2 The Effects of Image Sensor Choice On Phenotypical Analysis And Processing

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [14]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants [91]. Consequently, in low-cost settings, hyperspectral imaging of component-based phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [91]. Additionally, low-cost hardware has constrained bandwidth, storage, and compute power. Therefore, remote cloud image processing and analysis are required [92], which can be expensive [12].

2.3.2.3 Solutions, and Limitiations For Low-cost Large Data Throughput Image Processing Systems

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [12], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [11], and image compression techniques [16]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [17]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [14]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [16]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [18].

2.3.3 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Numerous opportunities exist to improve the overall affordability and generalisability of plant phenotyping in automated greenhouse environments. Low-cost plant phenotyping is not a nascent field of inquiry. However, many tracts of innovation have not been fully exploited. For instance, learning-based imaging techniques for analysis and processing can reduce analysis and segmentation costs but struggle when executed on resource-constrained hardware [69]. Therefore, there is a need for the development of efficient and optimised deep-learning or machine-learning models in resource-constrained environments. Similarly, many existing phenotyping systems are not generalizable to other plant species [14, 77] and rely on proprietary software or hardware [13]. Hence, there are opportunities to develop modifiable and flexible image-processing pipelines that are amenable to a broader variety of plant phenotyping contexts. Additionally, there are not many mobile applications for plant phenotyping. Mobile devices have advanced image sensors and can locally run image analysis and processing tasks, eliminating the need for cloud computing resources that increase phenotyping costs [18].

2.4 Conclusion

Ensuring an adequate food supply in response to the growing human population is crucial [23]. Image-based plant phenotyping has proven to be an effective tool for this purpose, aiding in the development of cultivars with maximum yield and resistance to various environmental stressors [31]. These methods are non-destructive, more time-efficient, replicable, and affordable than traditional methods [59]. However, image sensors in plant phenotyping generate large amounts of data, which necessitate robust image analysis and processing tools to handle this data effectively [90]. Large plant data throughput makes image-based plant phenotyping in automated greenhouse environments cost-prohibitive, as affordable off-the-shelf hardware and sensors often lack adequate local resources to store or process large amounts of data [91]. Consequently, remote cloud computing solutions handle image processing and analysis tasks [92], which can further increase operational costs in greenhouse management [12]. Image-data compression techniques, low-complexity image analysis and processing techniques, and low-cost off-the-shelf hardware are effective in reducing the cost of image-based plant phenotyping by reducing the computational requirements and data throughput [16, 11, 12]. However, these solutions come with marked trade-offs in terms of accuracy, precision, depth, and breadth of plant trait analysis. Moreover, existing solutions are not generalizable to other environments and plant species [6, 14]. In short, there is a need for low-cost, open-source, flexible, modifiable, and robust plant phenotyping image processing and analysis tools that are effective and efficient on resource-constrained hardware [12]. Future research should develop low-cost image processing tools that are precise, accurate, and adaptable to various plant monitoring contexts or parameters. In doing so, automated image-based techniques for plant monitoring will become more accessible in low-cost greenhouse environments.

Materials and Methods

3.1 Dataset

The PlantVillage dataset is used in our study. PlantVillage is a collection of 54309 annotated diseased and healthy leaf images spanning several plant species, including tomato [20]. The size of the subset of tomato leaf images is 18162. Several images of each tomato leaf were captured outside under natural light in various lighting conditions and orientations, against a black or grey background, using a standard digital camera [20, 1]. The images are captured in the RGB colour space and have dimensions 256x256. We have selected PlantVillage as our primary dataset because it is widely used in image processing research for plant disease detection [93, 94, 95, 96, 97, 98]. Using this dataset will allow future researchers to compare our proposed method to previous studies.

3.2 Image processing pipeline

Many studies in image-based plant disease detection typically have predetermined phases in their image processing pipeline. Namely, preprocessing, segmentation, and feature extraction [95, 99, 100, 101, 102]. We follow suit and include each of those phases in our image processing pipeline.

3.2.1 Preprocessing Phase

Preprocessing involves tasks such as noise removal, image enhancement, and colour space conversion. In image acquisition, noise from the imaging sensor, the environment, or another source can compromise the accuracy of downstream image processing and classification phases. For example, we choose a global thresholding segmentation method, which we discuss later, and these methods often perform poorly in the presence of noise or uneven illumination [103]. Nevertheless, preprocessing steps such as histogram equalisation and median filtering effectively mitigate noise and illumination issues [103, 104, 105]. Thus, preprocessing is a crucial phase of image processing pipelines, which enhances the overall accuracy of segmentation and image classification [1, 106]. Therefore, we apply several preprocessing steps to enhance the overall effectiveness and accuracy of our segmentation and classification phases.

We aim to compute texture features of the tomato leaf in each image using the grey-level co-occurrence matrix [107], which requires a grey-level image to calculate. Therefore, the first preprocessing step we perform is the conversion from the leaf image's original RGB colour space to the grayscale colour space.

Once we obtain the grayscale image of the tomato leaf, we apply histogram equalisation. This preprocessing method is used in the literature to add more contrast between the background and foreground of an image [104]. Image contrast enhancement is achieved by uniformly distributing the grey-level intensity values of the image [103], which makes histogram equalisation particularly useful for dealing with various illumination conditions [1].

Lastly, we apply median filtering to remove common salt-and-pepper noise from the image. Median filtering is an effective image enhancement preprocessing step that has been successfully used in plant leaf disease detection [79, 102]. Noise manifests as sharp transitions in image intensity seen in the grayscale image histogram [103], therefore noise can negatively affect the accuracy of global threshold segmentation methods, which we use in this study; therefore, to enhance segmentation accuracy and robustness, we apply median filtering to denoise the image.

3.2.2 Segmentation Phase

We use the Otsu method [108] for segmentation. The global thresholding Otsu method is used for segmentation. Global thresholding methods partition images into regions based on their intensity values using the image's grey-level histogram. For simple background removal, we select a suitable threshold T. If the intensity of the image at a point (x, y) exceeds T, then that part of the image is considered part of the foreground; otherwise, it is considered part of the background. If applied as a spatial-intensity transform, a binary mask is obtained where the intensity value of the foreground is set to 1, and the background is set to 0. Global thresholding performs poorly when there is low contrast between the background and foreground, large amounts of noise, and when the background intensity varies significantly across the image [109]. More accurate segmentation can be achieved using adaptive thresholding [110]. Another option is learning-based segmentation, but it requires substantial amounts of training data to achieve reliable segmentation [111]. However, global thresholding is a simple, easy to implement, computationally efficient [103, 110]. For this reason, we make the trade-off of poorer segmentation accuracy and robustness for computational efficiency. Moreover, global thresholding methods are applied in many studies in the literature for image-based plant phenotyping applications [93].

The Otsu method automatically selects a threshold value that maximises the interclass variance of the intensity levels between the background and foreground in the image's grey-level histogram. This method is widely used in the literature [93, 112], and can be utilised for background removal in plant image processing pipelines [100]. However, the Otsu method is sensitive to noise and lacks robustness to uneven illumination [93]. Despite the method's shortcomings, we have selected it because Otsu segmentation yields good segmentation results, is easy to implement, and is highly efficient [112, 93].

Concretely, for each leaf image, our segmentation phase aims to segment the leaf from its background. During feature extraction, we wish to obtain textural features of the leaf using information represented in the leaf's grayscale image. Therefore, background removal is crucial in ensuring that feature extraction considers the textural information of the leaf, rather than the background. The aforementioned Otsu segmentation method is applied to the grayscale leaf image obtained from the preprocessing phase. The result of the segmentation is a binary mask, and background removal is achieved by applying the binary mask to the preprocessed grayscale leaf image.

3.2.3 Feature Extraction Phase

After segmentation, we have the isolated leaf image in the grey-scale colour space, and proceed to feature extraction. We are only extracting textural features from metrics calculated using a grey-level co-occurrence matrix [107]. Textural features are empirically some of the best predictors of leaf disease detection [113, 114], since the occurrence of a plant disease induces specific changes in the texture of the leaf [115]. Thus, we choose textural features because we can construct feature vectors from these texture metrics to train a classifier to detect the presence or absence of a tomato leaf disease [116, 115, 114].

Gray-level co-occurrence matrices (GLCM) [107] measure the statistical spatial distribution of gray-level tones, and several parameters can be calculated from these matrices to investigate the textural properties of an object in an image [117]. GLCM derived textural features are widely used in the literature to detect plant leaf diseases [114, 115].

Following the approach in [115], we compute a GLCM from the grayscale leaf image produced by the segmentation phase. Thereafter, we extract the following parameters from the GLCM: energy, contrast, sum of squares, correlation, entropy, sum entropy, cluster shade, cluster prominence, and homogeneity metrics [114, 107, 115]. Together, all these calculated GLCM parameters form a feature vector, which we will use in our classification mechanism, as discussed in the next section.

3.3 Classification Mechanism

For our classification mechanism, we use a support vector machine (SVM) [19] to determine if a tomato leaf is diseased or healthy. SVM is a supervised machine learning method that has been shown to be an effective classifier of leaf diseases [115, 1, 118, 98]. SVM classifies several different classes by finding an optimal hyperplane that linearly separates them. SVMs can learn non-simple datasets and generalize well [1].

To obtain the training dataset, we apply our aforementioned image processing steps to obtain texture feature vectors. The feature vector is labeled as healthy if it is generated from a healthy PlantVillage tomato leaf image; otherwise, it is labeled as diseased.

With this training dataset, we train our Support Vector Machine using the radial basis function kernel [119]. To avoid overfitting and bias from the dataset, we perform 10-fold cross-validation [120]. In this training methodology, the data set is divided equally or nearly equally into 10 subsets. Then, cross-validation is performed 10 times, where one set is selected as the test set, and the union of the others forms the training set. The cross-validation procedure is repeated 10 times.

3.4 Analysis

We aim to measure the reliability of our binary SVM tomato classifier by evaluating its F-measure, accuracy, precision, actual positive rate, and true negative rate [120]. These metrics are commonly used to measure the performance of binary classifiers and predictive models [121]; therefore, their use allows comparison of our system with other studies. Additionally, we measure the average classification time of the system, which is the time it takes for classification to occur from the start of preprocessing to the end of classification.

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Plagiarism Pledge

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